

Week 2 Summary

Learn Quantum Mechanics Using a Quantum Computer

Quantum Coins — Superposition & Interference

Module: Superposition with Qiskit – IBM Quantum Learning

August 27, 2026

1. Overview

This module explores **superposition**—the ability of a quantum system to exist in a combination of classically allowed states until measured. Using Qiskit and the Aer simulator, we compared a classical coin flip with a quantum “coin flip” (Hadamard gate) and discovered that quantum superposition carries **phase** information, giving rise to **interference**—a phenomenon impossible in classical probability.

2. Classical vs. Quantum Coin

Classical coin: $S(\text{coin}) = \frac{1}{2}|\text{up}\rangle + \frac{1}{2}|\text{down}\rangle$ — coefficients *are* the probabilities.

Quantum coin (after Hadamard gate on $|0\rangle$):

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Here, the **square** of the (complex) amplitudes gives probabilities: $P(0) = P(1) = \frac{1}{2}$. Sampling 1000 shots with **Sampler** gives a $\sim 50/50$ histogram—identical to the classical case at first glance.

3. Phase & Interference

Quantum amplitudes are *complex*: $c_i = |c_i|e^{i\phi_i}$. The phase ϕ_i governs interference.

Double Hadamard ($H \cdot H|0\rangle$):

$$H\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) = \frac{1}{2}[(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle)] = |0\rangle$$

$|0\rangle$ terms add (*constructive*); $|1\rangle$ terms cancel (*destructive*).

H– $P(\pi)$ –H: Inserting a phase gate $P(\pi)$ flips the sign of $|1\rangle$, reversing the interference so the outcome is always $|1\rangle$.

4. Non-Classical Behaviour

Using **Estimator** to measure $\langle X \rangle$ on $H|0\rangle$:

$$\langle X \rangle \approx 1 \implies \text{always } |+\rangle, \text{ never } |-\rangle$$

The quantum coin is *random* along Z but *deterministic* along X —impossible for any classical probabilistic system.

5. $\sqrt{\text{NOT}}$ Gate (SX)

The SX gate rotates $|0\rangle$ by 90 around the x -axis (coin on its edge). Expectation values after SX:

$\langle X \rangle$	$\langle Y \rangle$	$\langle Z \rangle$
≈ 0	≈ -1	≈ 0

Random along X, Z ; deterministic along Y . A *coin sitting on its edge* is a much better analogy for superposition than a coin flipping in the air.

6. Bloch Sphere

Any pure single-qubit state is a point on the unit sphere:

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

All single-qubit gates are *rotations*: X (NOT) = 180 around x ; SX = 90 around x ; $P(\phi)$ = ϕ around z ; H = 90 around y then 180 around x .

7. Key Takeaways

- Superposition $\neq$ classical randomness.** Phase coherence enables interference; a coin flip does not.
- Phase is the secret ingredient.** The same gate can produce opposite outcomes depending on the input phase.
- Gates = rotations.** Deterministic & reversible. Randomness enters only at *measurement*.
- Bloch sphere** fully describes a single isolated qubit with angles (θ, ϕ) .